

Pen-and-Paper Exercises

k -Nearest Neighbours, K -Means and Linear Regression

No calculator needed, except where marked with ★.

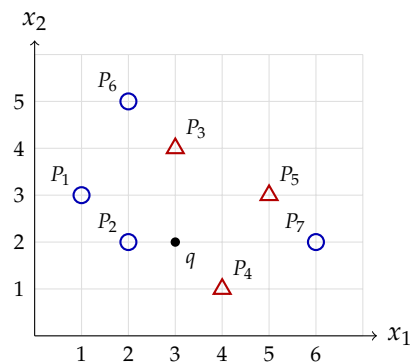
How to work. Show every intermediate step: distances, assignments, sums. A correct number with no working earns no credit in the exam. When a question asks for an interpretation, answer in two or three sentences and quote your numbers.

Part I k -Nearest Neighbours

Exercise 1 Classifying a point by hand

Seven labelled points in the plane belong to two classes, circles ($\circ$) and triangles ($\triangle$). We want to classify the query point $q = (3, 2)$.

Point	x_1	x_2	Class
P_1	1	3	$\circ$
P_2	2	2	$\circ$
P_3	3	4	$\triangle$
P_4	4	1	$\triangle$
P_5	5	3	$\triangle$
P_6	2	5	$\circ$
P_7	6	2	$\circ$



- Compute the Euclidean distance from q to each P_i (you may keep squared distances for the ranking) and sort the points from nearest to farthest.
- Predict the class of q with $k = 1$, $k = 3$, $k = 5$ and $k = 7$.
- What does the $k = 7$ classifier predict for *any* query point? Explain why a very large k underfits.
- Repeat the $k = 3$ prediction with the Manhattan distance $d_1(u, v) = |u_1 - v_1| + |u_2 - v_2|$. Does the prediction change?
- ★ In *distance-weighted k -NN*, each neighbour votes with weight $1/d(q, P_i)$. Give the prediction for $k = 5$ (Euclidean). Compare with (b).

Exercise 2 Choosing k by leave-one-out

A one-dimensional training set with two classes A and B :

x	1	2	4	5	8	9	11	15
class	A	A	A	B	B	A	B	B

In *leave-one-out* (LOO) validation, each point in turn is removed, classified by k -NN using the 7 remaining points, and compared with its true label.

- Compute the LOO error (number of mistakes out of 8) for $k = 1$. Present your work as a table: point, neighbour(s), prediction, correct or not.
- Same for $k = 3$ and $k = 5$.
- Which k would you choose? Point to the specific training points that make $k = 1$ perform badly and explain what they represent.

- (d) Why is it forbidden to include the point itself among its neighbours when computing this error? What would the error of 1-NN be if we did?

Exercise 3 Why features must be scaled

A bank describes customers by age (years) and annual income (euros). Query customer $q = (30, 40\,000)$. Two stored customers: $a = (32, 52\,000)$ and $b = (55, 41\,000)$.

- (a) Compute $d(q, a)$ and $d(q, b)$ in raw units. Which customer is nearer? Which feature decides?
- (b) On the training set, age has standard deviation 10 years and income has standard deviation 10 000 euros. Standardise the *differences* (divide each difference by its feature's standard deviation) and recompute both distances. Which customer is nearer now?
- (c) Would expressing income in thousands of euros instead of euros change the answer to (a)? What does this tell you about unscaled k -NN?
- (d) The standard deviations must be computed on the training set only. Explain why, and name the scikit-learn object that makes this automatic.

Exercise 4 The curse of dimensionality ★

Data points are spread uniformly in the unit hypercube $[0, 1]^d$. We want a small sub-cube around a query point that contains a fraction p of the data, i.e. a sub-cube of volume p .

- (a) Show that the side length of this sub-cube is $e_d(p) = p^{1/d}$.
- (b) Compute $e_d(0.01)$ for $d = 1, 2, 10, 100$.
- (c) Interpret: are the “1% nearest neighbours” in dimension 100 still *near*? What does this imply for k -NN on raw high-dimensional data?

Part II K-Means

Recall the two steps of Lloyd's algorithm: *assign* each point to its nearest centre, then *update* each centre to the mean of its points. The *inertia* is $J = \sum_i \|x_i - \mu_{c(i)}\|^2$.

Exercise 5 Running K-means by hand; the role of initialisation

One-dimensional data: 1, 2, 3, 8, 9, 10, 25. We use $K = 2$.

- (a) **Initialisation A:** $\mu_1 = 1, \mu_2 = 2$. Run Lloyd's algorithm until the assignments stop changing. At each iteration, write the decision threshold $(\mu_1 + \mu_2)/2$, the two clusters and the new centres.
- (b) Compute the final inertia for initialisation A.
- (c) **Initialisation B:** $\mu_1 = 1, \mu_2 = 25$. Run the algorithm to convergence and compute the final inertia.
- (d) Both runs have converged, yet the inertias differ. What does this tell you about Lloyd's algorithm? How does scikit-learn's `n_init` parameter address it?
- (e) Compute the inertia of the partition $\{1, 2, 3\}, \{8, 9, 10\}, \{25\}$ ($K = 3$). Given your results, discuss whether 25 is best seen as a cluster or as an outlier.

Exercise 6 k -means++ initialisation

With the same data 1, 2, 3, 8, 9, 10, 25, k -means++ picks the first centre uniformly at random; suppose it picks 1. Each next centre is drawn with probability proportional to $D(x)^2$, the squared distance from x to the nearest centre already chosen.

- (a) Compute $D(x)^2$ for every point and the probability of each being picked as the second centre.
- (b) What is the probability that the second centre is 25? Is this good or bad news for these data, given Exercise 5?
- (c) Explain in one sentence why k -means++ is sensitive to outliers.

Exercise 7 Why Lloyd's algorithm never increases the inertia

- (a) Let $x_1, \dots, x_m \in \mathbb{R}^p$ be the points of one cluster. Show that $f(\mu) = \sum_{j=1}^m \|x_j - \mu\|^2$ is minimised at $\mu = \bar{x} = \frac{1}{m} \sum_j x_j$. (Compute the gradient.)
- (b) Explain why the *assignment* step cannot increase J , and why the *update* step cannot increase J .
- (c) Deduce that the algorithm stops after a finite number of iterations. (Hint: how many partitions of n points into K groups exist?)

Part III Linear Regression

Throughout this part we use the five observations

x	1	2	3	4	5
y	2	3	5	4	6

and the model $\hat{y} = wx + b$ fitted by minimising the mean squared error $L(w, b) = \frac{1}{n} \sum_i (y_i - wx_i - b)^2$.

Exercise 8 The least-squares line by hand

- (a) Derive the closed-form solution by setting $\partial L / \partial b = 0$ and $\partial L / \partial w = 0$. Show that

$$b = \bar{y} - w\bar{x}, \quad w = \frac{S_{xy}}{S_{xx}}, \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

- (b) Compute $\bar{x}, \bar{y}, S_{xx}, S_{xy}$, then w and b .
- (c) Compute the fitted values, the residuals $e_i = y_i - \hat{y}_i$, and check that $\sum_i e_i = 0$. Explain why this always holds when the model has an intercept.
- (d) Compute $SSE = \sum e_i^2$, $SST = \sum (y_i - \bar{y})^2$ and $R^2 = 1 - SSE/SST$. Interpret R^2 in one sentence.
- (e) Predict y at $x = 6$. Would you trust a prediction at $x = 50$? Why?

Exercise 9 The normal equations

Write the model as $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\theta}$ with $\boldsymbol{\theta} = (b, w)^\top$ and a design matrix $\mathbf{X}$ whose first column is all ones.

- (a) Write $\mathbf{X}$ and compute $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$.
- (b) Solve $\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y}$ by inverting the 2×2 matrix. Check that you recover Exercise 8.
- (c) Suppose we add a third column equal to $2x$. What happens to $\mathbf{X}^\top \mathbf{X}$? What does this mean for the solution, and how does Ridge regression fix it?

Exercise 10 One step of gradient descent

- (a) Show that $\frac{\partial L}{\partial w} = -\frac{2}{n} \sum_i x_i (y_i - \hat{y}_i)$ and $\frac{\partial L}{\partial b} = -\frac{2}{n} \sum_i (y_i - \hat{y}_i)$.
- (b) Start from $w = 0, b = 0$. Compute $L(0, 0)$ and both partial derivatives. (Use $\sum x_i y_i = 69, \sum y_i = 20, \sum y_i^2 = 90$.)
- (c) Perform one update with learning rate $\eta = 0.05$: $w \leftarrow w - \eta \partial L / \partial w, b \leftarrow b - \eta \partial L / \partial b$.
- (d) ★ Compute L after the step. Did the loss decrease? Compare the new (w, b) with the optimum $(0.9, 1.3)$: which parameter overshoot, and why is that not a contradiction?

Exercise 11 Ridge regression in one dimension

Centre the data ($\tilde{x}_i = x_i - \bar{x}, \tilde{y}_i = y_i - \bar{y}$) and fit $\tilde{y} = w\tilde{x}$ by minimising $\sum_i (\tilde{y}_i - w\tilde{x}_i)^2 + \lambda w^2$.

- (a) Show that $w_\lambda = \frac{S_{xy}}{S_{xx} + \lambda}$.
- (b) Compute w_λ for $\lambda = 0, 5, 10, 90$ and the intercept $b_\lambda = \bar{y} - w_\lambda \bar{x}$ for $\lambda = 5$.
- (c) What happens as $\lambda \rightarrow \infty$? Why is the intercept usually *not* penalised?
- (d) Why must features be standardised before applying Ridge with several features?

Exercise 12 Outliers and leverage

Add one observation to the five points and refit by least squares.

- (a) Case 1: add $(x, y) = (3, 14)$. Compute the new $\bar{x}, \bar{y}, S_{xx}, S_{xy}$, then w and b .
- (b) Case 2: instead add $(x, y) = (9, 4)$. Same computation.
- (c) Both new points lie far from the original line. Compare the effect on the slope and explain the difference using the notion of *leverage* (distance of x_i from $\bar{x}$).
- (d) Name one loss function that is less sensitive to such points than the squared error.